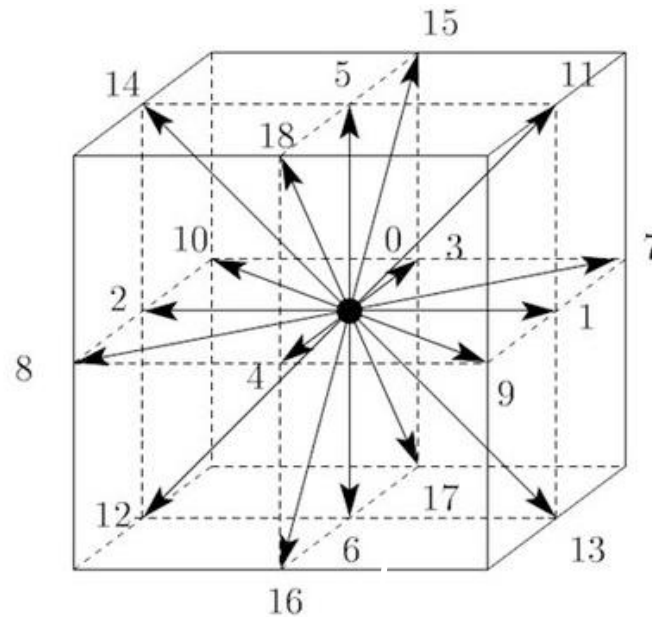
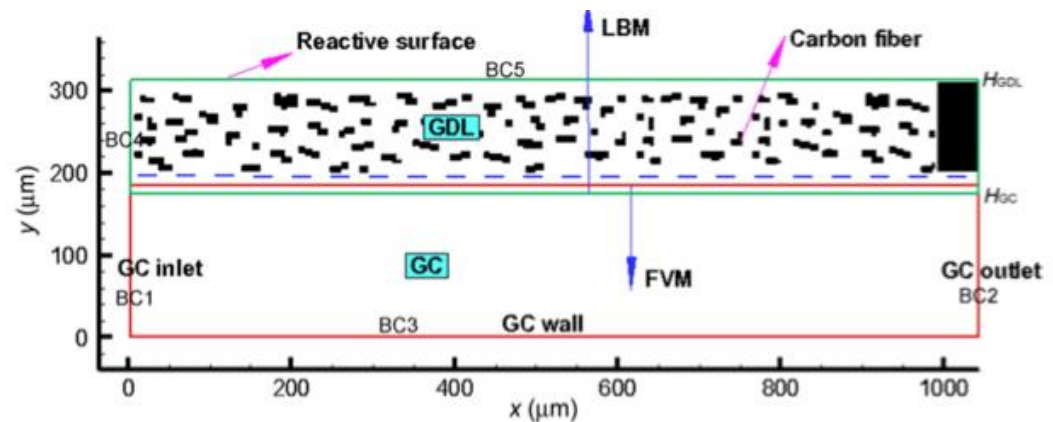


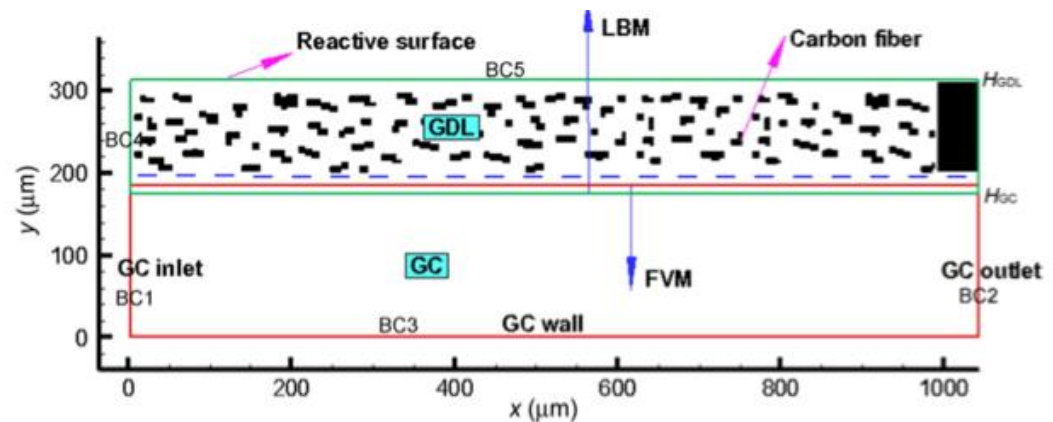


Lattice Boltzmann Method

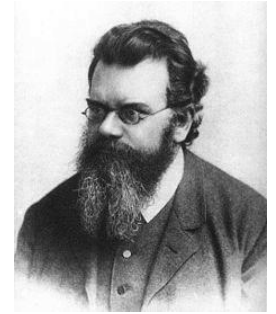
Dr. Alessandro De Rosis

Slides adapted from Dr. Alistair Revell and Dr. Joseph O'Connor





The Boltzmann Equation



- **Boltzmann (1844-1906): Austrian physicist**
- **First to realise connection between micro and macro scale**
 - Entropy connected with number of states of the system
 - Number of microscopic states for a given energy far greater for macroscopic equilibrium
 - For example, the arrangement of particles in a litre container of gas
- **Derived the distribution function at the same time as Maxwell**
 - All ideal gases have a specific equilibrium distribution function
- **But also proposed a framework for how the system returns to equilibrium**
 - The Boltzmann Equation
 - Cornerstone of the lattice Boltzmann method

The Boltzmann Equation

- **Consider particles at $f(r, c, t)$ to which a force is applied**
 - c increases to $c + Fdt$
 - r increases to $r + cdt$
- **Consider all particles in the interval $drdc$ before and after the force**
 - If there are no collisions between these particles there is no change in the number
 - However, if there are collisions (likely) we need to recompute the number in this interval
- **The collision operator is the rate of change of collisions:**

$$f(r + cdt, c + Fdt, t + dt)drdc - f(r, c, t)drdc = \Omega(f)drdc dt$$

- Which (in absence of external force) can be shown to be simplified as:

$$\frac{\partial f}{\partial t} + c \cdot \nabla f = \Omega$$

- Macroscopic quantities are recovered via summation:

$$\rho(r, t) = \int m f(r, c, t) dc \qquad \rho(r, t) u(r, t) = \int m c f(r, c, t) dc$$

BGK Operator

- **Boltzmann equation hard to solve since collision term is very complicated**
 - Integro-differential equation since collision term itself is a function of f

- **In 1954, Bhatnagar, Gross and Krook (BGK) introduced a simplified model for collision operator:**

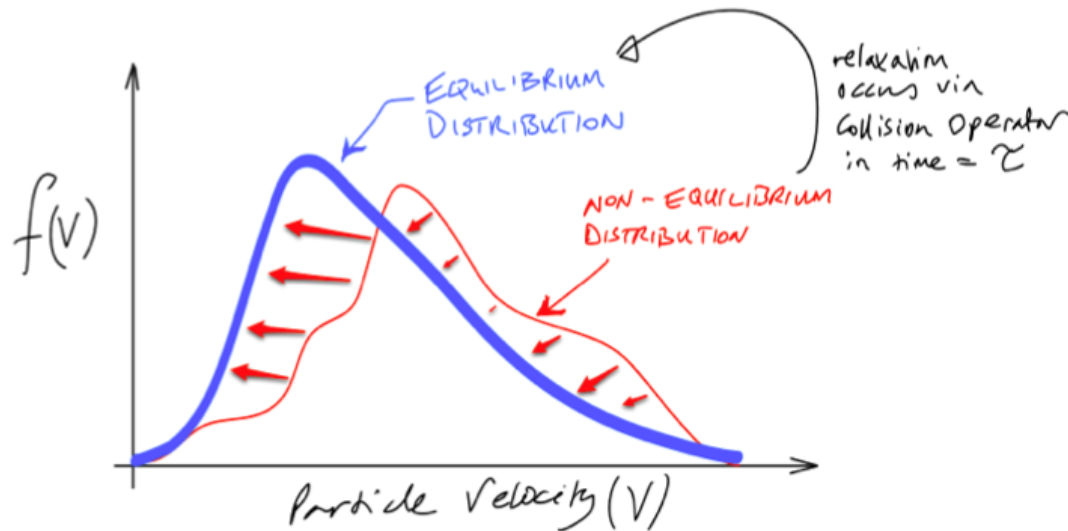
$$\Omega = \omega(f^{\text{eq}} - f) = \frac{1}{\tau}(f^{\text{eq}} - f) \quad \omega = 1/\tau$$

- Where f^{eq} denotes the Maxwell-Boltzmann equilibrium function
 - This was a key step in applying the LBM to real-world engineering problems
 - BGK is the work horse of the lattice Boltzmann method
- **It refers to the time taken for a given distribution function f to relax back to the equilibrium state**

Time Relaxation to Equilibrium

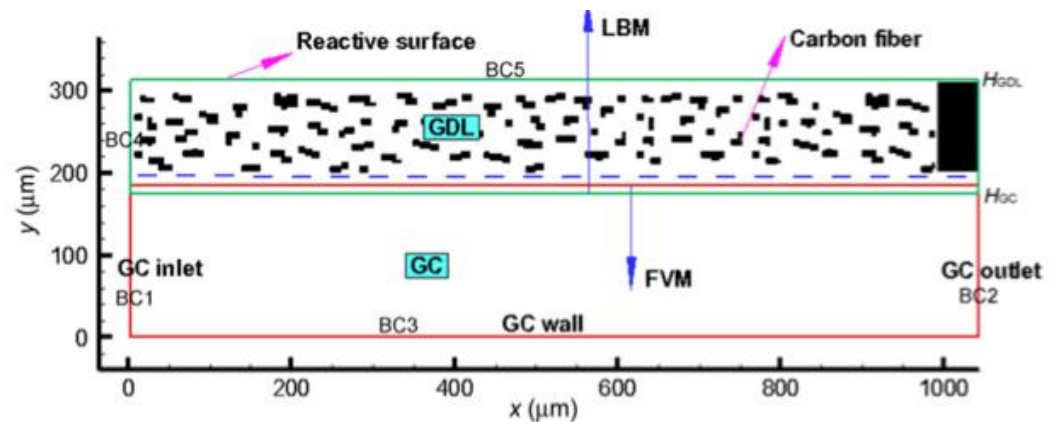
- **BGK approach assumes distribution function relaxes back in time τ**

- Can be shown to be proportional to the viscosity of the fluid
- In many ways the collision operator acts as a diffusion term



- **More complex time relaxation schemes exist**

- Twin Relation Time (TRT): for two component flow (multiphase)
- Multiple Relaxation Time (MRT): necessary for turbulent flow



Discretisation of the LBM

- The equation for the LBM has been defined, we now discretise for use in a numerical framework

$$\frac{\partial f}{\partial t} + c \cdot \nabla f = \frac{1}{\tau} (f^{\text{eq}} - f) \qquad f_i(r + c_i \Delta t, t + \Delta t) = f_i(r, t) + \frac{\Delta t}{\tau} [f_i^{\text{eq}}(r, t) - f_i(r, t)]$$

- **Note**

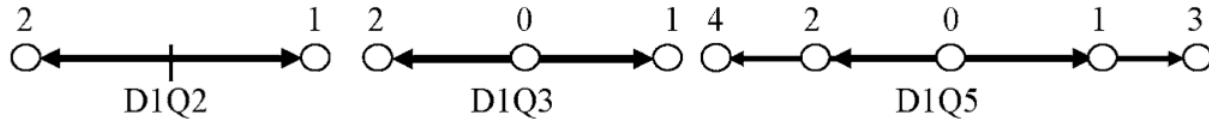
- That Δt is generally 1 (we nondimensionalise to lattice units)
- The equation is a linear partial differential equation
- The equation looks like an advection equation with a source term
- The left-hand side of the equation represents the advection (streaming)
- The right-hand side term represents the collision process (source term)

- **We need to divide the domain into discrete points: the lattice**

Lattice Models

- The common terminology used in the LBM is to refer to the dimension of the problem and the number of velocities using **DnQm**
 - **n** refers to the dimension of the model
 - **m** refers to the number of discrete velocities for the model

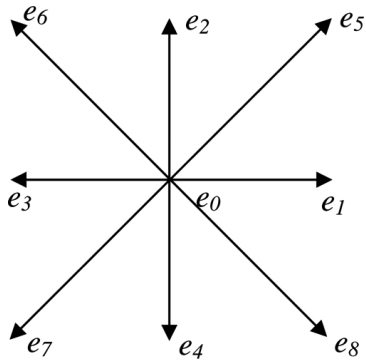
- In 1D you can have D1Q2, D1Q3, D1Q5



- For D1Q3 there are three velocity vectors
 - $C_1 = 0$
 - $C_2 = 1$
 - $C_3 = -1$
- Each component has a variable contribution dependent on the model
 - This is defined using weights, which are computed using Gaussian quadrature

Lattice Models

- In two dimensions the most common model is D2Q9:

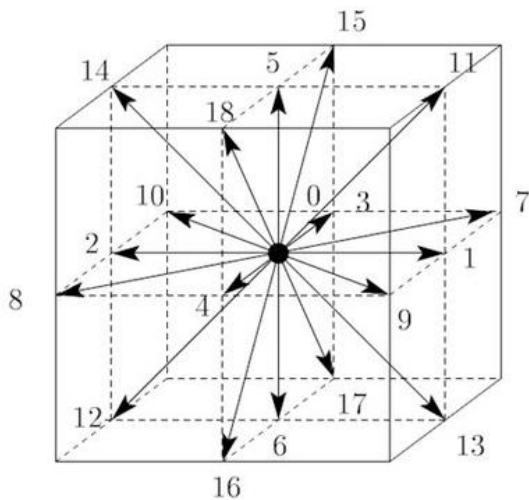


$$w_0 = \frac{4}{9}$$

$$w_{1-4} = \frac{1}{9}$$

$$w_{5-8} = \frac{1}{36}$$

- In three dimensions the most common is D3Q19:

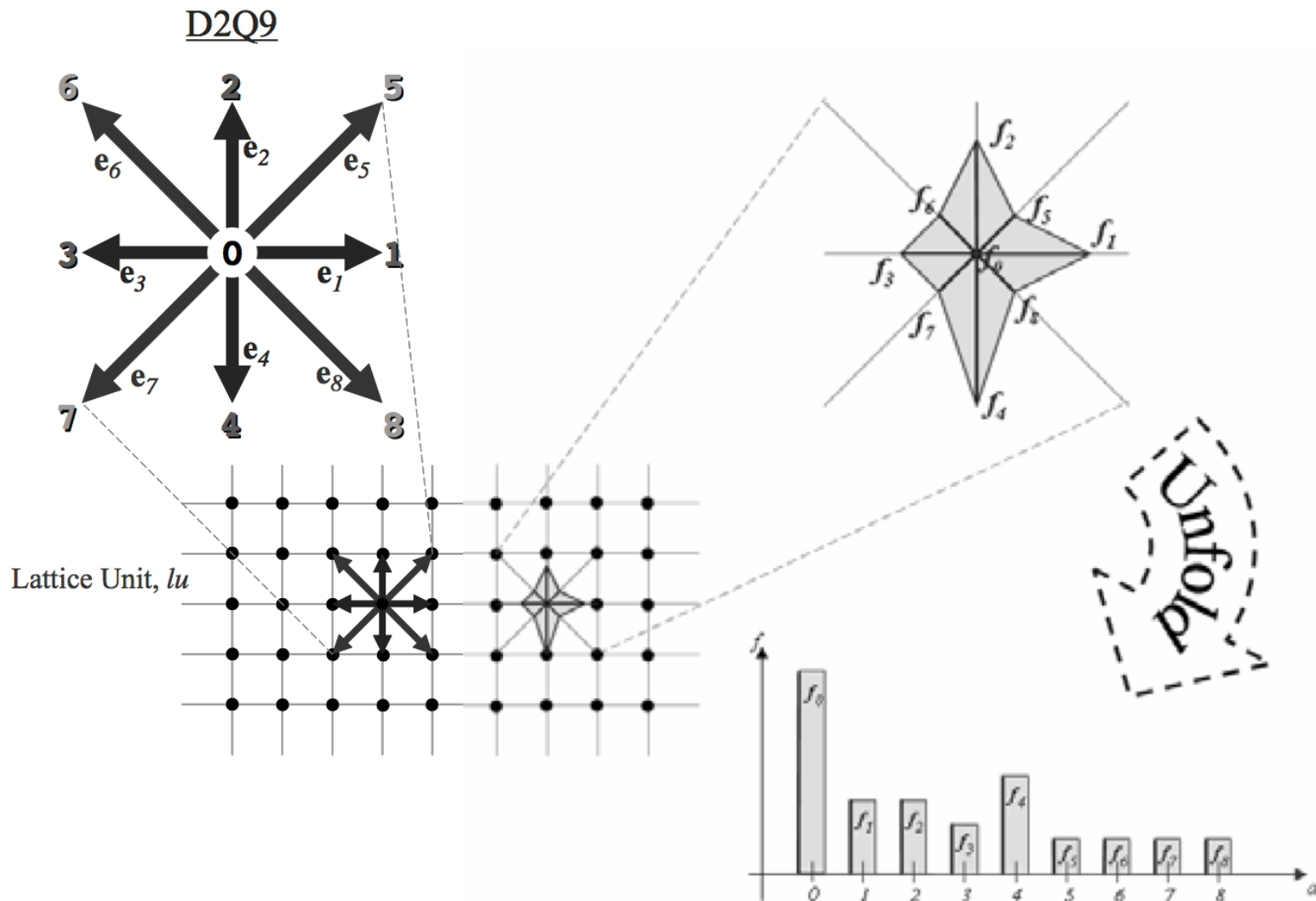


$$w_0 = \frac{12}{36}$$

$$w_{1-6} = \frac{2}{36}$$

$$w_{7-18} = \frac{1}{36}$$

Discretisation of LBM-BGK



Validity of LBM for Different Flows

$$f(x, c, t) = f^N(x, c, t) = w(c) \sum_{n=0}^N \frac{1}{n!} a^n(x, t) \quad a^{(n)}(x, c, t) = \int f(x, c, t) H^{(n)}(c) dc$$

■ Hermite polynomials

- Based on 13-moment theory of Grad (1945), the distribution function may be expanded over velocity space using orthonormal Hermite polynomials
- Coefficients of Hermite polynomials are identical to the macroscopic variables

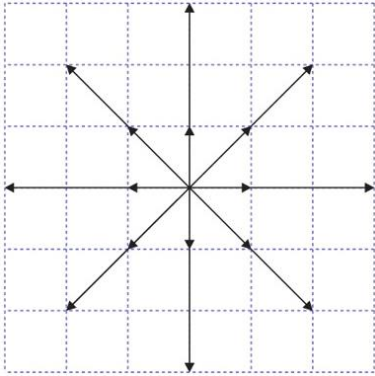
$$a^{(0)} = \int f dc = r, \quad a^{(1)} = \int cf dc = ru$$

- Burnett equations: rarefied gas or nanofluidics

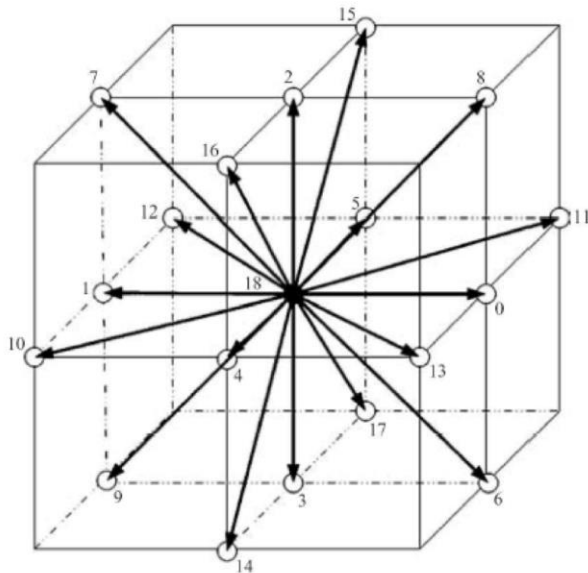
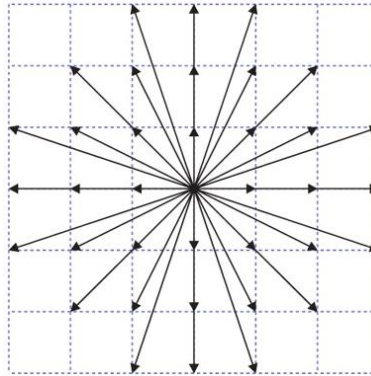
H(n)	Resulting model	NS momentum only	Burnett	NS+energy	Suitable physics
4	D2Q9 D3Q19	OK at small Ma	Not valid	Not valid	Incompressible non thermal flow
6	D2Q17 D3Q39	OK	OK at small Ma	OK at small temp variation	Supersonic regime
8	D2Q37 D3Q121	OK	OK	OK	All!

Higher Order Discretisations

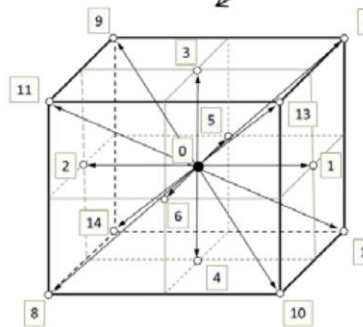
D2Q17



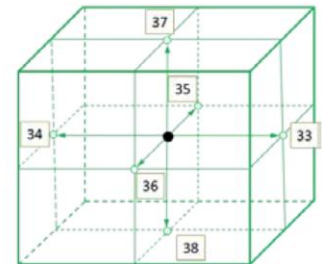
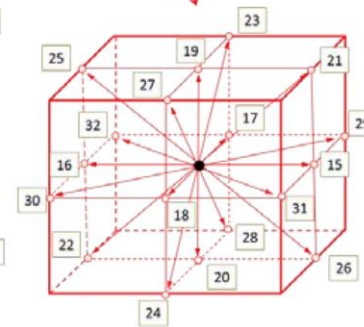
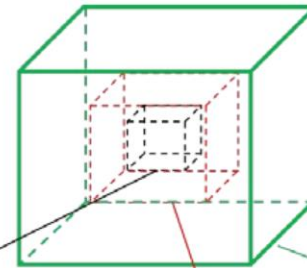
D2Q37



D3Q19 Model
(freestream $Ma_{\max} \sim 0.4$)



D3Q39 Model
(freestream $Ma_{\max} \sim 3.0$)



Discussion

- What are the implications of going for 'higher order'?

Equilibrium Distribution Function

- **Recall, equilibrium distribution in D dimensions is:** $f^{eq}(v) = \rho \left(\frac{m}{2\pi k_B T} \right)^{D/2} e^{-\frac{mv^2}{2k_B T}}$

- **Which can be reformulated:**

$$f^{eq}(v) = \frac{\rho}{(2\pi c_s^2)^{D/2}} e^{-\frac{(\mathbf{c}-\mathbf{u})^2}{2c_s^2}}$$

- Noting that the speed of sound is: $c_s = \sqrt{k_B T/m} = 1/\sqrt{3}$
- Noting that the relative velocity is given by $\mathbf{v} = \mathbf{c} - \mathbf{u}$

- **This can be approximated using Taylor expansion up to 2nd order to obtain:**

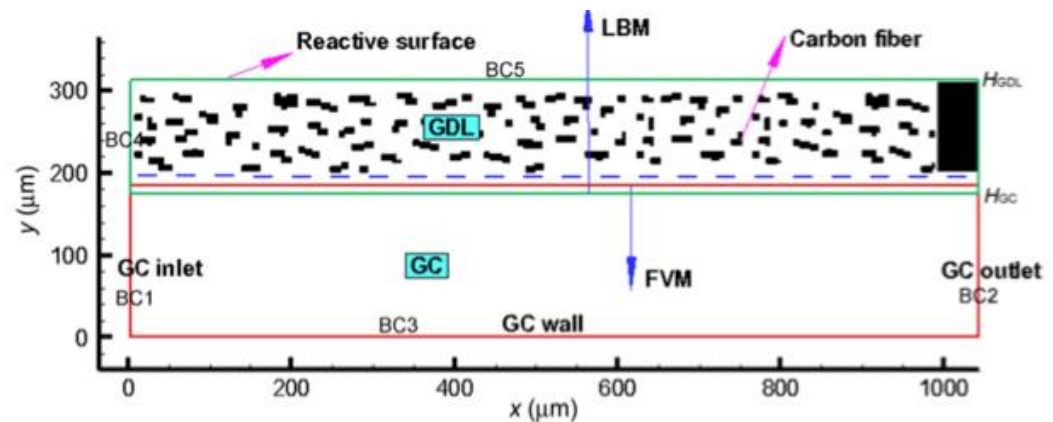
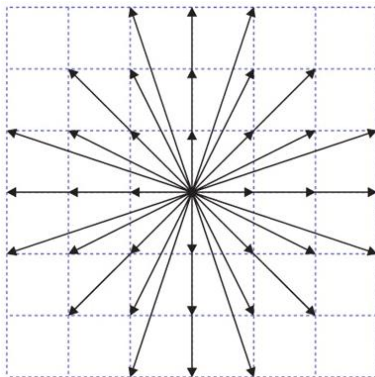
$$f^{eq}(v) = \frac{\rho}{(2\pi c_s^2)^{D/2}} e^{-\frac{c^2}{2c_s^2}} \left[1 + \frac{\mathbf{c} \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c} \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right]$$

- **Finally discretised into the component distributions with relevant weightings:**

$$f_i^{eq} = \rho w_i \left[1 + 3(\mathbf{c}_i \cdot \mathbf{u}) + \frac{9}{2}(\mathbf{c}_i \cdot \mathbf{u})^2 - \frac{3}{2}(\mathbf{u} \cdot \mathbf{u}) \right]$$

Part 2: Implementation

- The Boltzmann Transport equation
- Discretisation
- Comparison with FVM



Comparison of FVM vs LBM

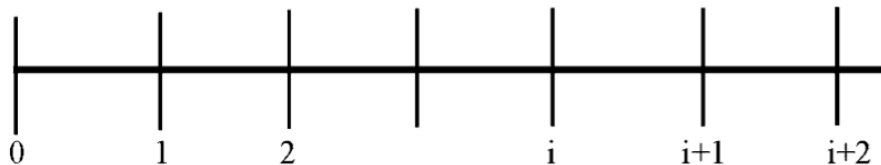
- **Consider a 1D diffusion equation:**

- The dependent variable diffuses in an infinite medium in both directions
- Diffusion speeds up as parameter alpha increases

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$\frac{1}{\tau} \approx \alpha \frac{1}{\ell^2}$$

- **First consider using finite differences:**



$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2}$$

- **Can be written as:**

$$T_i^{n+1} = T_i^n(1 - \omega) + \omega T_i^{\text{eq}}$$

$$\omega = \frac{2\alpha\Delta t}{\Delta x^2}$$

Comparison of FVM vs LBM

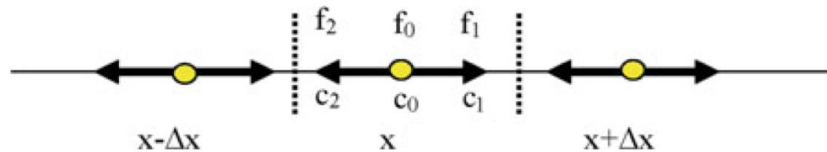
■ Consider a 1D diffusion equation:

- The dependent variable diffuses in an infinite medium in both directions
- Diffusion speeds up as parameter alpha increases

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$\frac{1}{\tau} \approx \alpha \frac{1}{\ell^2}$$

■ Now for the lattice Boltzmann method:



$$\frac{\partial f_k(x, t)}{\partial t} + c_k \cdot \frac{\partial f_k(x, t)}{\partial x} = \Omega_k$$

$$c_k = \frac{\Delta x}{\Delta t}$$

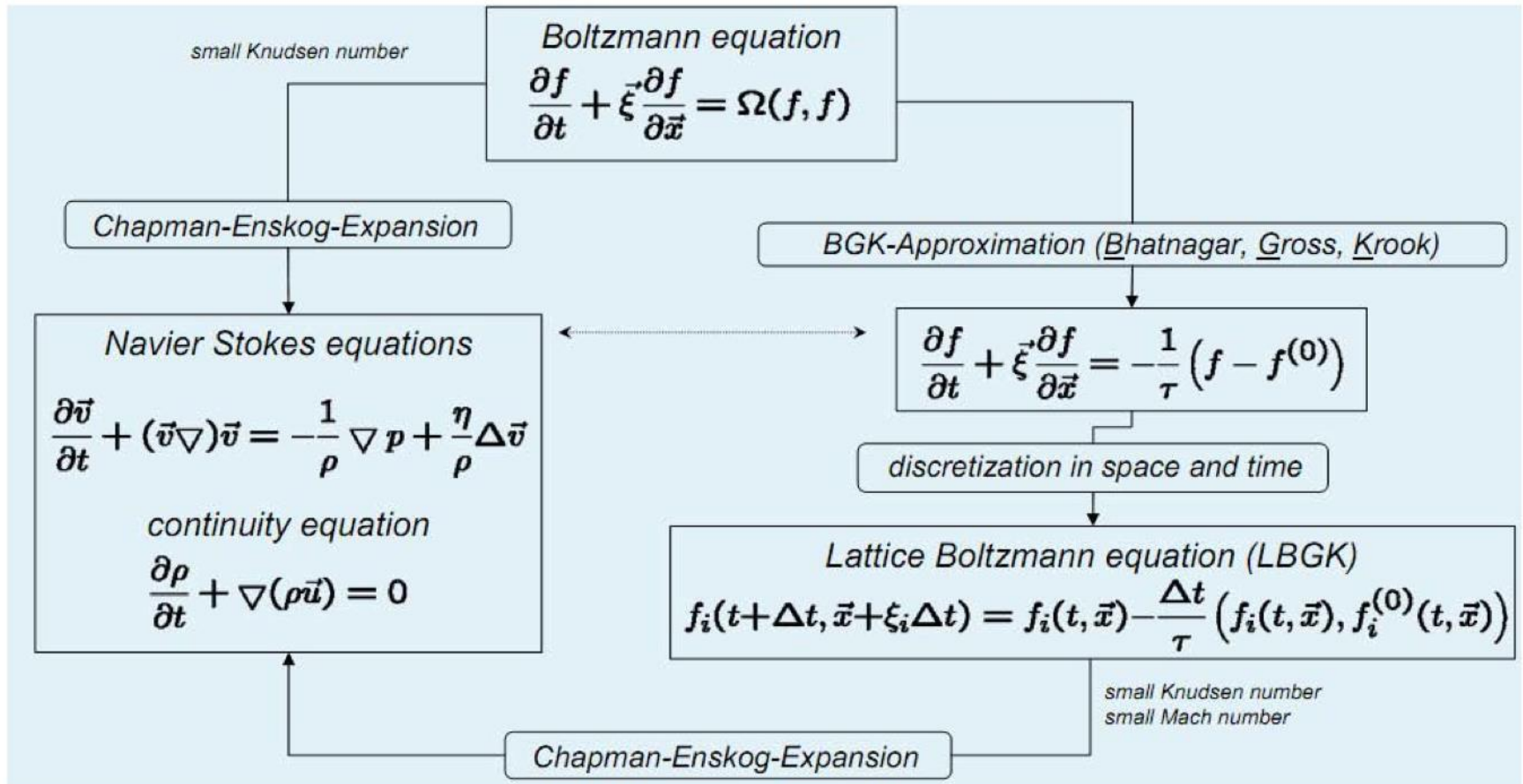
$$\Omega_k = -\frac{1}{\tau} [f_k(x, t) - f_k^{\text{eq}}(x, t)]$$

■ Can be written as:

$$f_k(x + \Delta x, t + \Delta t) = f_k(x, t)[1 - \omega] + \omega f_k^{\text{eq}}(x, t)$$

$$\omega = \frac{\Delta t}{\tau}$$

Overview: LBM to NS

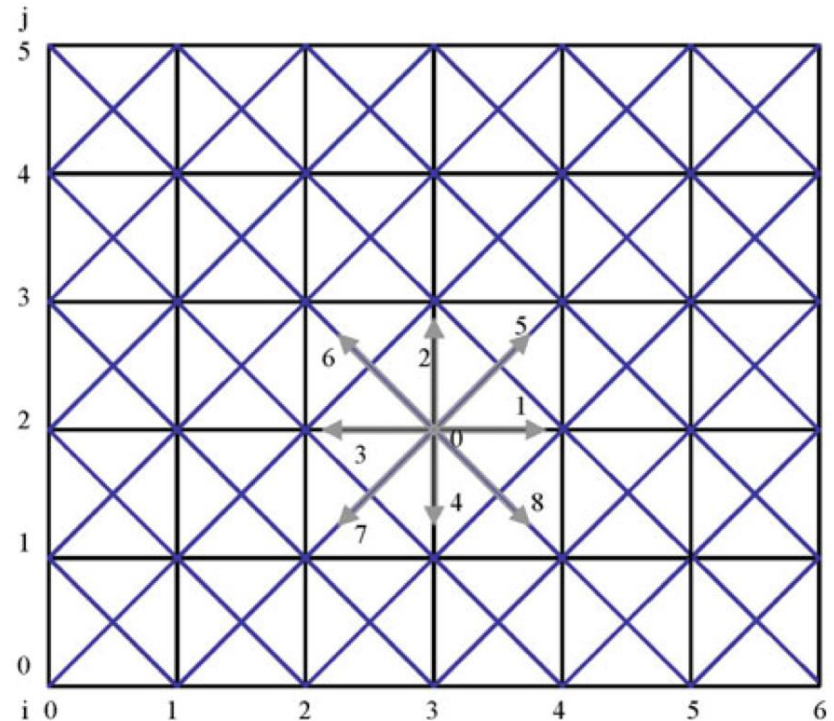


Algorithm

■ The algorithm for LBM can now be defined

- We consider boundary conditions later

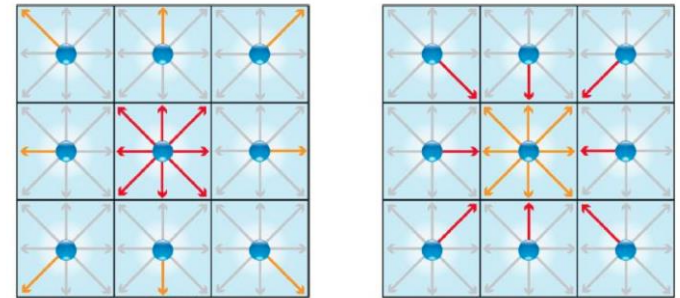
1. Initialise
2. Compute equilibrium function
3. Collide
4. Stream
5. Boundary conditions
6. Compute macroscopic values
7. Loop back to Step 2



The Actual Code

■ Compute equilibrium function

```
for(int dir = 0; dir < 9; ++dir) {
    pop_eq[dir] = weight[dir] * den * (1 + 3 * (cx[dir]*vel_x + cy[dir]*vel_y)
        + 4.5 * SQ( cx[dir]*vel_x + cy[dir]*vel_y) - 1.5 * (SQ(vel_x) + SQ(vel_y)));
}
```



■ Stream & Collide

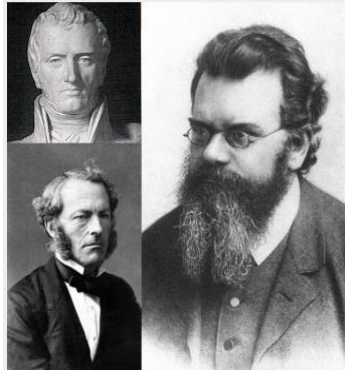
```
for(int dir = 0; dir < 9; ++dir) {
    Xnew=X+cx[dir];
    Ynew=Y+cy[dir];
    if(Xnew!=-1 && Xnew!=Nx && Ynew!=-1 && Ynew!=Ny){
        pop [dir][Xnew][Ynew] = pop_old[dir][X][Y] * (1 - omega)
            + pop_eq[dir] * omega + force_latt[dir];
    }
}
```

■ Compute macroscopic values

```
for(int dir = 0; dir < 9; ++dir) {
    rhosum += pop[dir][X][Y];
    usum += cx[dir]*pop[dir][X][Y];
    vsum += cy[dir]*pop[dir][X][Y];
}
```

Navier Stokes vs Boltzmann

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}$$



$$\frac{\partial f}{\partial t} + c \cdot \nabla f = \frac{1}{\tau} (f^{eq} - f)$$

■ Navier Stokes

- 2nd order PDE
- Nonlinear: $(\mathbf{u} \cdot \nabla) \mathbf{u}$
- Pressure via Poisson equation
- Nonlocal (elliptic)

■ Lattice Boltzmann Method

- 1st order PDE
- Linear
- No pressure
- Local

Navier Stokes vs Boltzmann

For LBM:

- Easy to do complex geometry
- Very simple to program
- Extremely parallelisable

Against LBM:

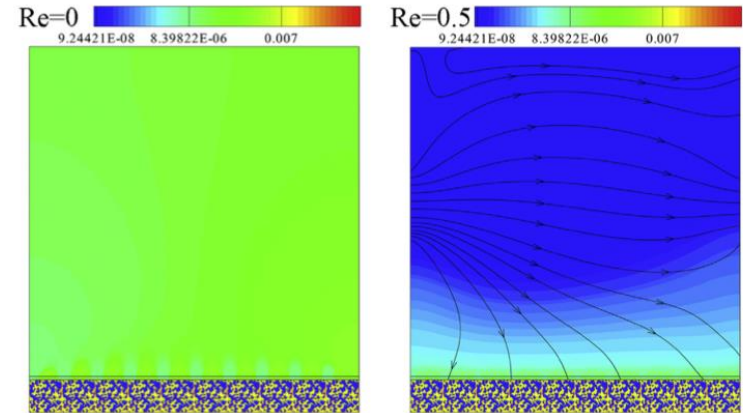
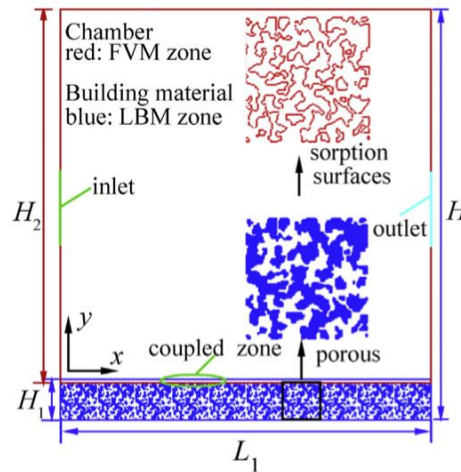
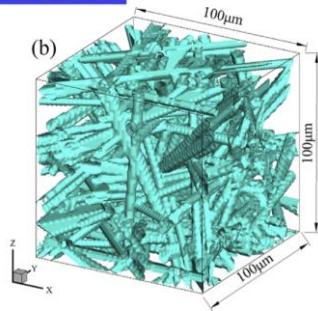
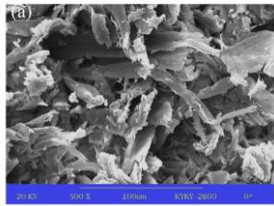
- Regular grids are a limitation
- Subject to instabilities with low viscosity
- Limited to low Reynolds and Mach numbers

Conclusion: *it's 'horses for courses'*

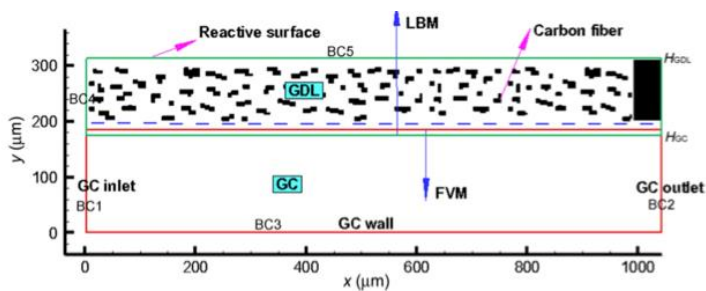
- LBM most valuable in low Re with complex geometry
- FVM still relevant in high Re limit

Best of Both (NS + LBM)

- Increasingly we see CFD involving both FVM and LBM



Y.-T. Mu et al. / Building and Environment 92 (2015) 236–245



L. Chen et al. / International Journal of Heat and Mass Transfer 55 (2012) 3834–3848

